Validation report

Voice Postcode Assistant · NAIC 2026 · PS2 Voice-First Access · civic information · Live: https://muideen27--voice-postcode-web.modal.run

N-ATLaS is an initiative of the Federal Ministry of Communications, Innovation and Digital Economy, and powered by Awarri Technologies.

All numbers in this report come from the service's own anonymised interaction log, exported with `scripts/export_log.py --pull --since 2026-10-11T00:00:00+01:00` and filled in by `scripts/fill_submission.py`. Developer test turns are never counted (they are logged to a separate folder, and the export starts at the pilot start time).

1. Method

- **Pilot dates:** 11–12 October 2026 (first logged turn 2026-10-11T05:57:07+01:00, last 2026-10-12T11:37:46+01:00).
- **Testers:** 15 people (7 browser sessions; one phone was shared by several testers), all non-developers. Recruitment: Personal WhatsApp invitations to friends, family and colleagues, plus fellow NAIC participants; live, supervised sessions on 11–12 October 2026. Testers used their own phones, and one phone was shared. No names, phone numbers or NIN were collected.
- **Shared phone:** one session on 12 Oct (08:07–08:53 WAT, 36 logged turns in all four languages) was one phone used by several different people, so the session count (7) is lower than the number of people.
- **Fellow participants:** some testers on 12 Oct were fellow NAIC participants; they could not be separated in the anonymised log.
- **Developer sessions excluded:** the developer's own test sessions and device are left out of every number (salted session hashes listed in `scripts/excluded_hashes.txt`).
- **Setting:** testers used their own Android phones and mobile data, opening the live link. A facilitator read the tester script (`docs/pilot_checklist.md`, section 4) and did not touch the phone. During sessions both GPUs were kept warm with `scripts/warm.sh`.
- **What a tester did:** pick a language → read the welcome and consent → answer "Did you know your new postcode before today?" (yes / no / skip) → get a postcode by sharing location, ask by voice, check a code, ask a question → answer "Did this help?" after each reply.
- **Consent:** shown in the chosen language before anything is stored. Declining deletes the session; only an anonymous daily count of declines is kept.
- **Privacy:** no names, phone numbers or NIN are asked for. Location is used for one lookup and never stored. Users appear only as salted one-way hashes. Voice recordings are deleted after transcription. Stored text is scrubbed of numbers, phones and emails and capped at 300 characters. No coordinates, audio or raw session IDs are in the log.

2. Headline numbers

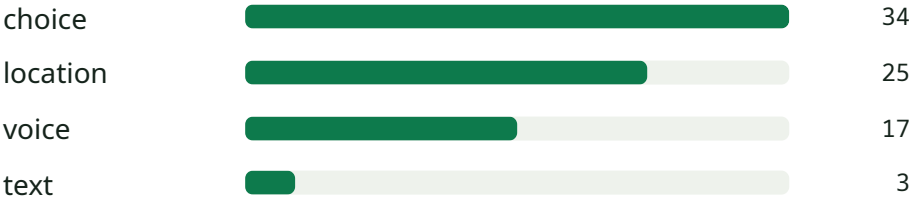
Measure	Value
Real interactions (logged turns after consent; PS2 minimum 50)	**79**
Content turns (voice, text, location)	45
Unique users (browser sessions; a phone shared by several people counts once)	7
Turns that reached the Postcode API / with a postcode	22 / 0
"Did this help?" yes / no	6 / 1
Knew their postcode before: yes / no / skipped or not asked	1 / 4 / 2
Declined consent (anonymous count)	0

Interactions by language



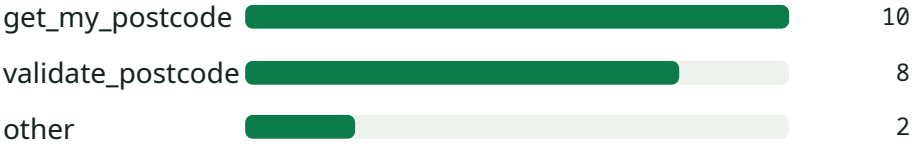
Value	Count	Share
en	28	35%
yo	22	28%
ha	15	19%
ig	14	18%

Input type



Value	Count	Share
choice	34	43%
location	25	32%
voice	17	22%
text	3	4%

What people asked for (voice and text turns)



Value	Count	Share
get_my_postcode	10	50%
validate_postcode	8	40%
other	2	10%

3. Findings

Did people know their postcode before? Asked once, right after consent and **before any postcode is shown**, so the answer is not biased by the result: 4 said no, 1 said yes, and 2 skipped or were not asked.

Feedback. After each answer the app asks "Did this help?": 6 yes and 1 no.

Latency per stage (milliseconds; voice = speech recognition, llm = intent, api = Postcode API):

Stage	Turns	p50	p95
asr	17	1751	20634
llm	20	2676	4565
api	22	124	320
total	79	309	7832

Declines per day:

Date (WAT)	Declines
—	0

Example transcripts

At most three real transcripts, each checked by hand to contain no personal data (no names, phone numbers, addresses or other identifying details). All other evidence in this report is aggregated.

Language, input	What the speech model heard (exactly as transcribed)	Approximate meaning (developer's translation)	Routed to
English, voice	"hi, good morning. so i want to check for my post code. can you help me do that please?"	(English)	ask for location
Yorùbá, voice	"ẹjọ́rọ̀ mú fẹ́ mú pósíkòdì ilé mi, ìjọsẹ̀ ríi bá mi ṣe àyẹ̀wò pósíkòdì ilé mi ẹjọ́."	"Please, I want my house postcode; please help me check my house postcode."	ask for location
Igbo, voice	"biko onu, i nyere ha m aka lelee koodu zipam"	"Please, help me check my postcode."	ask for the code or location

4. Official postcode data during the pilot

The service checks one known postcode (`LA-11-A12-GN-01`, valid on 3 Oct 2026) at every web start and every 30 minutes, and records the result per day:

Date (WAT)	valid	not_found	auth	error	Last result	Official data available?
2026-10-09	0	0	29	0	auth	unknown
2026-10-10	0	20	35	0	not_found	no (official data returned nothing)
2026-10-11	0	55	0	0	not_found	no (official data returned nothing)
2026-10-12	0	26	0	0	not_found	no (official data returned nothing)

5. Limitations (stated plainly)

1. **The official test data has been empty since at least 9 Oct 2026.** Evidence: - 3 Oct 2026: a live reverse lookup of 6.6018, 3.3515 (Ikeja) returned `LA-11-A12-GN-01`, and a lookup of that code returned `valid` (recorded responses in `tests/fixtures/postcode/`). - 9 Oct 2026: with a valid key, reverse lookups at nine places (Ikeja ×2, Victoria Island, Yaba, Abuja Wuse 2, Abuja Garki, Kano, Enugu, Ibadan) all returned "no postcode within range", the same code returned `not_found`, and the API's own autocomplete for "LA11" returned no suggestions. - From 10 Oct 2026 the canary above records the daily result (`not_found` = valid key, no data). NIPOST has been informed. During these days the app never invents a postcode. A location lookup answers honestly: "The official postcode database has no postcode for this spot yet". For a **code check** (typed or spoken) the app uses the canary: when the lookup says `not_found` and today's canary is not `valid`, it does not tell the user the code is wrong. It replies "{code} has the right format, but the official postcode database can't confirm it today. Here is what each part means:" followed by the meaning of each part (state, Local Government Area, district, area, unit), and logs `api_status = unconfirmed_db_down` with `postcode_found` left empty. When the canary is `valid`, a `not_found` code is reported as not found. In the numbers above, unconfirmed code checks are therefore neither "valid" nor "not found".
2. **Igbo speech recognition is labelled beta** in the app; Igbo users are offered "Type instead".
3. **Languages not yet reviewed by native speakers:** Yorùbá, Hausa and Igbo. Testers saw the developer's draft translations; the native-speaker review was not completed before the pilot (English is final).
4. **Speech recognition accuracy was measured offline only** (recorded test clips), not live: the pilot does not store audio, so live word accuracy cannot be computed.
5. Location answers depend on the phone's GPS accuracy and on the API's 50 m / 250 m search radius. **Repeated attempts are counted as logged:** on 12 Oct one user shared their location four times within 20 seconds (11:28–11:29 WAT); each attempt is a separate logged turn and is included in the totals, as is every other turn. Interactions are counted exactly as logged, without merging repeats.
6. Language-model cold start (about 2 minutes) is avoided during sessions by keeping the GPUs warm; outside sessions the first request may wait and the page says so.